

Jerell James

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Software engineer with eight years of experience in a small specialist software team within an engineering consultancy. Built complex simulation, geospatial and internal business products using TypeScript, React, Rust, Zig, C#/.NET, WebAssembly, WebGPU and Azure.

Experience

SOFTWARE ENGINEER | PACE CCS | JAN 2018 - AUG 2026

Research and development of engineering software for carbon capture, thermodynamics, geospatial network modelling, and internal business operations using Rust, Zig, TypeScript, C#/.NET, Julia, WebAssembly, WebGPU, and Azure.

- *flōwer* - Rust, Zig, TypeScript, React, Electron, Bun, WebAssembly, WebGPU, ONNX
 - Independently built a geospatial flow-network platform as an internal tool for constructing, querying, evaluating, and simulating directed-graph pipeline networks. Supported more efficient delivery of engineering projects by reducing the time required to evaluate alternative designs.
 - Implemented both a Zig/WebAssembly prototype and a newer Rust/WebAssembly core for TOML network loading, hierarchical property inheritance, query execution, shapefile handling, coordinate reprojection, and OLGA .key import/export.
 - Designed GPU-accelerated steady-state simulation workflows using ONNX thermodynamic property surfaces, WebGPU compute shaders, and Zarr-backed result storage for pressure-enthalpy design-space exploration.
- Digital Twin - C#/.NET, TypeScript, React, ONNX, Azure
 - Built the initial steady-state flow-network digital twin application, including the front-end and substantial contributions to the solver, which iteratively resolves fluid properties to satisfy downstream-equipment constraints.
 - Delivered digital twins to external clients, providing interactive views of the calculations underpinning engineering project reports.
- *Dim* - Zig, Rust
 - Independently built a dimensional analysis and unit-conversion library with compile-time dimensional safety, derived-unit arithmetic, SI/Imperial/CGS registries, affine units, formatting, and a CLI expression evaluator.
 - Compiled the library to WebAssembly for browser integration, exposing unit-safe calculations to engineering applications such as flōwer.
- Phase Envelope Generator - TypeScript, Next.js, React, WebGPU, Three.js, ONNX Runtime Web
 - Independently developed an interactive thermodynamic phase-envelope and flash-calculation web application for multi-component fluid compositions.
 - Integrated ONNX Runtime Web inference for phase, density, entropy, temperature, viscosity, and related thermodynamic properties, with worker-based batch evaluation for responsive plotting.
- Timesheets - TypeScript, React, Next.js, NestJS, Azure, Docker, Terraform, GitHub Actions
 - Built and maintained the company-wide project, time submission, and reporting system used by employees to record project work and generate progress/cost reports.

- Designed and implemented its event-sourced NestJS backend, including CQRS-style commands and events, read projections, reindexing tools, Azure Table Storage, Azure AD authentication, and queue-backed projection updates.
- [Ichthys LNG Phase 2a FEED](#)
 - Optimised subsea design configurations using an evolutionary algorithm; paper presented at the BHR 19th International Conference on Multiphase Production Technology, Cannes, France, June 2019.

Education

- 2021, BSc in Digital & Technology Solutions, Open University
- 2020, Foundation Degree in Data Analytics and Innovation, Open University
- 2017, 4 AS/A-Levels A*-C, Esher College